Roll No.: _____

Amrita Vishwa Vidyapeetham

Amrita School of Artificial Intelligence, Faridabad

B.Tech Mid-Term Examinations – March 2025

Second Semester

Artificial Intelligence & Data Science

23PHY114 Computational Mechanics 2

Duration: 2 hours | Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	This course introduces students to the concepts and techniques of computational applied Mechanics, providing them with a solid foundation in analyzing and solving mechanical engineering problems using computational methods.
CO02	This course equips students with the necessary skills to apply the Finite Element Method (FEM) and Computational Fluid Mechanics (CFD) for the analysis of common robotic applications.
CO03	This course familiarizes students with fundamentals of Hilbert spaces and quantum mechanics for computational applications.
CO04	This course familiarizes students with the basic robotic control and its implementation for simple robotic systems.

Answer all questions

Short-Answer Questions

1. Define stress and strain. How do tensile, compressive, and shear stresses fundamentally differ in their effect on materials? Discuss the Hooke's Law and stress-strain analysis beyond the elastic limit with reference to the stress-strain curve for low-carbon steel.

(4 marks)

A metallic rod of length 1 m and cross-sectional area 2 cm^2 is subjected to a tensile force of 10 kN. The material initially obeys Hooke's law with Young's modulus = 200 GPa. However, after a strain of 0.002, the modulus drops to 150 GPa due to microstructural changes. What will be the total elongation if the same force is applied beyond the 0.002 strain limit?

(1 mark)

2. State any four general principles of stress-strain analysis. What is the fundamental aim of stress analysis for a solid state system? What is a Stress Matrix and what do each of its elements represent? How do normal and shear strain differ in their interpretation?

(5 marks)

3. What is Poisson's ratio, and what is its significance when its value is 0 or 0.5?

(3 marks)

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Explain the underlying motivation for the form of the three-dimensional Hooke's Law in the context of material behavior under stress.

(2 marks)

4. How does the stress matrix transform under a coordinate transformation? Discuss the cases when the Mohr circle be just a point and when would it just touch the origin,

(3 marks)

A mild steel block is subjected to the following stress conditions: Normal tensile stress along x-axis: $\sigma_x = 100$ MPa, normal tensile stress along y-axis: $\sigma_y = 50$ Mpa, shear stress: $\tau_{xy} = 30$ MPa. A thin layer of paint is applied to the surface. The paint has tensile strength: 120 MPa and adhesion strength: 40 MPa.

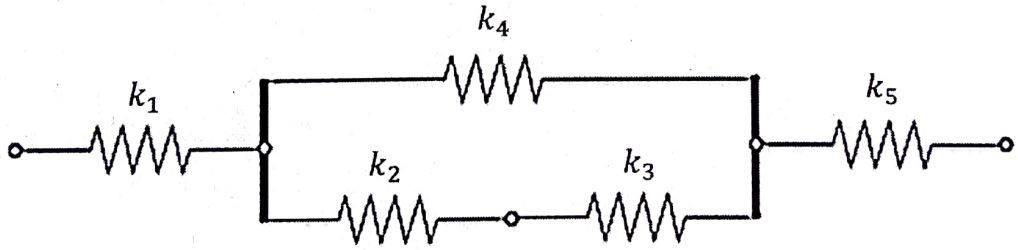
Based on Mohr's Circle analysis, will the paint crack first due to normal stress or peel off?

(2 marks)

5. Derive the 1D governing equation of elasticity. Use finite difference method to find its finite-difference form. Given any two kinds of linear elements in Finite Element Analysis (FEA) and what kind of deformative tendencies do they have?

(5 marks)

6. What is the algorithm (fundamental steps) for the Finite Element Method (FEM)? Give two of the conditions considered in the direct stiffness method for FEA. What is the stiffness matrix for the following combination of springs?



(5 marks)

Long-Answer Questions

7. What is a functional? Give an example of a functional in the context of mechanics. Define the variation of a functional and explain its physical significance with a diagram.

(3 marks)

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Define the Gâteaux derivative. How is the Gâteaux derivative used to find stationary points of a functional? Consider the functional $J[y] = \int_0^{2\pi} (1 + y'(x)^2) dx$ with $y(0) = 0$ and $y(2\pi) = 2\pi$, with $y_1(x) = x$ and $\eta(x) = \sin x$, then what is the Gâteaux derivative of this functional?

(3 marks)

Write the total potential energy functional for a cantilever beam of length L and flexural rigidity EI. Explain how the *Principle of Minimum Potential Energy* leads to equilibrium conditions.

(2 marks)

Using the *Rayleigh-Ritz method*, approximate the deflection $u(x)$ using an algebraic approximation.

(2 marks)

8. Explain the motivation behind the Weighted Residual Method (WRM) in solving differential equations numerically – why is it necessary to enforce residual minimization rather than solving the equation directly?

(1 mark)

Given an equation of the form $\mathcal{L}(\psi) + f \neq 0$, derive the stiffness equation of the form: $KC = f$.

(3 marks)

What is the Galerkin Method? Use the Galerkin Method with trigonometric approximations to find the solutions for the beam subjected to moments:

$$EI\frac{d^2y}{dx^2} - M_0 = 0,\ x \in [0, L]$$

with boundary conditions $y(0) = 0,\ y(L) = 0$.

(3 marks)

Suppose we had a Modified Galerkin Method, where $w_i = N_i^2$, how would this affect the choice of a trigonometric approximation for the displacement in a beam subjected to moments?

Note: $\int_0^L \sin^3\left(\frac{\pi x}{L}\right) dx = \frac{1}{4}\int_0^L \left[3\sin\left(\frac{\pi x}{L}\right) - \sin\left(\frac{3\pi x}{L}\right)\right] dx = \frac{4L}{3\pi}$

(2 marks)

Given that the inner product $\langle f, g \rangle$ for two square-integrable real-valued functions f and g, in a general vector space, on an interval $[a, b]$ is

$$\langle f, g \rangle = \int_a^b f(x)g(x)dx$$

and we assume that the trial functions, weights and residuals are square-integrable real-valued functions, what can we say regarding the relation between the trial functions, weight functions and the residual, when considering the Galerkin method?

(1 mark)